

1.2) The difference between Free Software and Open Source Software:

The difference between free software and open source software is philosophical and ideological.

1. Free Software

Free software emphasizes user freedom.

The term comes from the Free Software Foundation (FSF), founded by Richard Stallman.

Software is considered "free" if it guarantees four fundamental freedoms:

1. Freedom to use the software for any purpose
2. Freedom to study how it works (access to the source code)
3. Freedom to modify the software
4. Freedom to redistribute copies (modified or unmodified)

"Free" means freedom, not gratuitous.

Examples of Free Software Licenses

- GPL (GNU General Public License)
- LGPL
- AGPL

The opposite of **free software** is **proprietary software**.

Before discussing the distinction between free software and open-source software, it's important to understand what proprietary software is.

Proprietary software is software whose source code belongs exclusively to a company or developer, who controls its use, modification, and distribution.

To better understand this, imagine a cake:

- Proprietary software is like buying a ready-made cake. You can eat it, but you don't know the ingredients or the recipe. You can't add more chocolate or reduce the sugar; you're entirely dependent on the baker.

- Free software, on the other hand, gives you the recipe. You see the ingredients, you understand the preparation, you can adapt the recipe to your taste, and even share your new version with your friends. The recipe is the source code; having it means maintaining control. Not having it means being completely dependent on whoever created it.

2. Open Source Software

Open source software emphasizes practical and technical advantages:

- Code quality
- Security
- Collaboration
- Performance

The source code is accessible, modifiable, and redistributable, but the approach is more pragmatic than activist.

Examples of Open Source Licenses

- MIT
- BSD
- Apache
- GPL (yes, it's also open source)

3. Essential Difference:

Aspect	Free Software	Open Source
Vision	Ethical and social	Technical and pragmatic
Objective	Defend user freedom	Produce better software
User	Freedoms Fundamental	Often implicit
Key Term	Freedom	Efficiency
Origin	FSF (Stallman)	Open Source Initiative

Note: All free software is open source.

However, not all open source software is necessarily considered "free" according to the FSF, especially if certain freedoms are limited.

Concrete example:

Linux: free and open source

Firefox: free and open source

Android: open source, but not entirely free (proprietary components)

Summary:

Free software: priority given to user freedom (use, study, modify, share).

Open source: priority given to software quality and efficiency thanks to open code.

The difference is primarily philosophical, not technical.

***All free software has open code,
But some open source software does not respect all the freedoms of free software.***

Example:

Imagine a book:

- Free software: you can:
 - Read the book
 - Copy the book
 - Modify the content
 - Distribute your version

Freedom is a fundamental right

- Open-source software: you can:
 - Read the book
 - Improve the content

The goal is to have a better book

Conclusion:

Same open code, but different philosophies

Free software → ethics and freedom

Open source → technology and performance